

Sakshi Arya

Curriculum Vitae

Case Western Reserve University
Department of Mathematics, Applied Mathematics and Statistics
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🌐 <https://sakshiarya.github.io/>

Research Interests

Multi-armed bandit problem, sequential decision making, nonparametric estimation, statistical learning theory, statistical inverse problems, spatio-temporal analysis

Employment

- July 2023 - Present **Assistant Professor**, *Department of Mathematics, Applied Mathematics and Statistics*, Case Western Reserve University
- Aug 2021- June 2023 **Eberly Postdoctoral fellow**, *Department of Statistics*, Penn State University, with Prof. Bharath Sriperumbudur
- Feb 2021- Aug 2021 **Postdoctoral Associate**, *School of Statistics*, University of Minnesota, with Prof. Snigdhanu Chatterjee
- July 2020- Jan 2021 **Postdoctoral scholar**, *Biological Sciences Division, Department of Public Health Sciences*, University of Chicago

Education

- 2014-2020 **PhD, Statistics**, *The University of Minnesota*, Twin-Cities, MN., Advisor: Prof. Yuhong Yang
- May 2014 **Masters of Science, Mathematics**, *Tata Institute of Fundamental Research, Center for Applicable Mathematics*, Bangalore, Karnataka, India
- May 2012 **Bachelor of Science, Mathematics**, *University of Delhi*, Lady Shri Ram College, India

Publications

- Sakshi Arya "Batched Nonparametric Bandits via k-Nearest Neighbor UCB" *Transactions of Machine Learning Research*, 2025
- Anuj Abhishek and Sakshi Arya. "Adaptive Estimation of a Function from its Exponential Radon Transform in Presence of Noise" *Sankhya A*, 2023
- Sakshi Arya and Yuhong Yang. "To update or not to update? Delayed Nonparametric Bandits with Randomized Allocation" *Stat*, 2021
- Sakshi Arya and Yuhong Yang. "Randomized Allocation with Nonparametric Estimation for a Contextual Multi-Armed Banit Problem with Delayed Rewards." *Statistics and Probability letters*, 2020

Zack W. Almquist, Sakshi Arya, Li Zeng and Emma S. Spiro. "Unbiased Sampling of Users from (Online) Activity Data." *Field Methods*, 2018

Under revision

Note: * indicates student co-authors under my supervision.

Sakshi Arya, Wentao Lin* "Non-Linear Model-Based Sequential Decision-Making in Agriculture" 2025

Sakshi Arya, Hyebin Song "Semi-Parametric Batched Global Multi-Armed Bandits with Covariates" 2025

Sakshi Arya, Bharath Sriperumbudur. "Kernel ϵ -greedy for Multi-Armed Bandits with Covariates" 2024 +

Lindsey Dietz, Sakshi Arya, Snigdhasu Chatterjee, Auroop Ganguly, Vishal Subedi "Predictions of damages from Atlantic tropical cyclones: a hierarchical Bayesian study on extremes" 2024 +

Sakshi Arya, Yuhong Yang. "Finite-Time Regret Bounds for Contextual Bandits with Delayed Rewards using Randomized Allocation" 2024 +

Working papers

Satarupa Bhattacharjee, Sakshi Arya, Bharath Sriperumbudur. "Kernel Single-Index Bandits: Estimation, Inference, and Learning"

Anuj Abhishek, Madhu Gupta and Sakshi Arya. "Fast and accurate Bayesian inference for non-linear inverse problems: An application to Electrical Impedance Tomography and Quantitative Photoacoustic Tomography."

Teaching

Case Western Reserve University

Fall 2025 **Instructor**, for STAT 301 (Introduction to Probability for Statistics)

Spring 2025 **Instructor**, for STAT 244 (Mathematical Statistics I)

Fall 2024 **Instructor**, for STAT 526 (Statistical Learning Theory)

Fall 2024 **Instructor**, for STAT 243 (Mathematical Statistics II)

Spring 2024 **Instructor**, for STAT 244 (Mathematical Statistics II)

Fall 2023 **Instructor**, for STAT 243 (Mathematical Statistics I)

Penn State University

Fall 2021 **Instructor**, for STAT/MATH 414 (Introduction to Probability Theory)

Fall 2022 **Instructor**, for STAT/MATH 414 (Introduction to Probability Theory)

University of Minnesota

Fall 2019 **Teaching assistant**, for STAT 5601 (Nonparametric methods)

Summer 2019 **Teaching assistant**, for qualifying exam training for PhD students

Spring 2019 **Instructor**, for STAT 4101 (Theory of Statistics I)

Fall 2018 **Instructor**, for STAT 3021 (Introduction to Probability and Statistics)

Summer 2017 **Instructor**, for STAT 3011 (Introduction to Statistical Analysis)

- Spring 2018

2014–2015 **Teaching Assistant STAT**

- 3011: Intro to Statistical Analysis (2)
- 3022: Data Analysis (1)
- 5701: Statistical Computing (1)
- 8052: Design of Experiments (1)

Talks at Conferences

Invited Talks

- “Structured Learning with Batched Bandits: From k-Nearest Neighbors to Single-Index Models” at the Data Science Seminar, Old Dominion University, October 2025.
- “Structured Learning with Batched Bandits: From k-Nearest Neighbors to Single-Index Models”, Seminar, Data Science Department, IISER Pune, India, July 2025.
- “Bayesian Hierarchical Modeling for Extremes” at the International Indian Statistical Association Conference, University of Nebraska-Lincoln, June 2025.
- “Single-Index Batched Bandits” at the International Indian Statistical Association Conference, Kochi, India, 2024.
- “Single-Index Batched Bandits” at the Joint Meeting of NZMS, AustMS and AMS, Auckland, New Zealand, 2024.
- “Predictions of damages from Atlantic Tropical Cyclones: a Hierarchical Bayesian study on extremes” The International Environmetrics Society Conference (TIES 2024), The University of South Australia, Adelaide, South Australia.
- “Kernelized epsilon-greedy algorithm for nonparametric bandits with covariates” 7th International Conference on Econometrics and Statistics (EcoSta 2024), Beijing Normal University, China.
- “Kernel Epsilon-Greedy Algorithm for Multi-Armed Bandits with Covariates” at CMStatistics, December 2023, Berlin, Germany.
- “Kernel Epsilon-Greedy Algorithm for Contextual Bandits” at the Computing and Uncertainty Analysis Seminar, Case Western Reserve University, November 2023.
- “Epsilon-Greedy Approach for Nonparametric Bandits” at JSM 2023, Toronto, Canada, August 2023.
- “Kernel Epsilon-Greedy Algorithm for Contextual Bandits” at International Indian Statistical Association Annual conference, Colorado School of Mines, June 2023.
- “Kernel Epsilon-Greedy Algorithm for Contextual Bandits” at Stochastic Modeling and Computation Statistics Seminar series, Penn State University, April 2023.
- “Kernel Epsilon-Greedy Algorithm for Contextual Bandits” at the department of statistics, University of Connecticut, October 2022.
- “Bayesian Hierarchical Modeling of Tropical Storms and their Damage Potential: A data driven study” at SIAM student chapter, University of Texas at Arlington, April 2022.
- “Applications of Adaptive Estimation in Nonparametric Regression” at Stochastic Modeling and Computation Statistics Seminar series, Penn State University, October 2021.
- “Adaptive estimation of function from exponential radon transform data” at SIAM Southeastern Atlantic Section Conference, Auburn University, September 2021.
- “Multi-Armed Bandits for Personalized Medicine” at University of St. Thomas, Minneapolis, MN, October 2019.

Contributed talks

- “Semi-Parametric Batched Global Multi-Armed Bandits with Covariates” at the Joint Statistical Meetings, Nashville, Tennessee, August 2025.
- “Kernel Epsilon-Greedy Algorithm for Contextual Bandits” at the 50th anniversary celebration of School of Statistics, University of Minnesota, October 2022.
- “Kernel Epsilon-Greedy Algorithm for Contextual Bandits” at Women in Statistics and Data Science (WSDS), St. Louis, Missouri, October, 2022.
- “Bayesian Hierarchical Modeling of Tropical Storms and their Damage Potential: A data driven study” at a contributed speed session in Joint Statistical Meeting (JSM), Washington DC, 2022.
- “Non-parametric Multi-Armed Bandits with Covariates and Delayed Rewards” at a contributed session in Joint Statistical Meeting (JSM), Denver, CO, 2019.
- “Multi-armed bandits with Covariates and Delayed Rewards” at the 7th International Workshop on Sequential Methodologies, Binghamton, NY, 2019.
- “Treatment Allocation using Multi-Armed Bandits with Covariates” American Statistical Association Twin Cities, Spring Meeting 2018
- “Geo-Social Rankings Derived from Activity based Apps” at workshop on Activity based Networks in SocInfo 2016.

Posters

- Machine Learning Summer School (MLSS, July 2019, University College London)
- IRSA Conference on “Causal Inference and Data Science” (Spring 2019, University of Minnesota)
- IRSA Conference “Cook’s Distance and Beyond” (Spring 2019, University of Minnesota)
- IMA Workshop on “Statistical and Computational Challenges in Precision Medicine” (Fall 2018, University of Minnesota)
- Seminar on Stochastic Processes (SSP), Brown University (2018, Providence, RI)
- International Indian Statistical Association (IISA) International Conference on Statistics (2017, Hyderabad, India)
- ASA Twin Cities, Fall Research Conference (2016, Mounds View, MN)

Non-Research Presentations

- Co-presented (with Haema Nilakanta) “*Shine with Shiny: An Introduction to R Shiny*” at the Business Analytics Symposium, Iowa State University, 2019.
- Co-organized (with Haema Nilakanta) a workshop on R Shiny during the University of Minnesota “Day of Data,” 2019.

Awards and Fellowships

2025–2026 Veale Faculty Fellow, Veale Institute for Entrepreneurship, Case Western Reserve University

University of Minnesota

Summer 2019 Martin Statistics Travel Award
 Summer 2018 Graduate Research Partnership Program Fellowship
 Summer 2017 Martin-Buehler Summer Fellowship Award
 Summer 2016 Lynn Lin Fellowship in Statistics

Travel Awards

Fall 2018 Women in Statistics and Data Science Conference, Cincinnati, OH
 Spring 2018 Seminar on Stochastic Processes, Brown University
 Fall 2017 International Indian Statistical Association Conference, Hyderabad, India

Other Honors

Summer 2019 Poster Award (Second Place), Machine Learning Summer School, London

Professional Involvement and Leadership

- Spring 2020 **Student Representative, Faculty Search**, *School of Statistics, University of Minnesota*, Coordinated student meetings and provided feedback to the selection committee.
- 2018–2019 **President**, *Association for India's Development, Minnesota Chapter (AID Minnesota)*
- 2017–2018 **Overall Project Coordinator**, *Association for India's Development (AID Minnesota)*
- Spring 2016 – Fall 2016 **Research Assistant**, *For Prof. Zack Almquist, Project: Activity-Based Networks*, University of Minnesota
- Summer 2016 **Statistics Consultant**, *Department of Speech-Language-Hearing Sciences, University of Minnesota*
- Summer 2015 **Statistics in the Community (STATCOM) Consultant**, *Compassionate Action for Animals (CAA)*, Investigated effects of CAA's Veg Fest.

Professional Service and Outreach

Advising

- Independent Study course: Wentao Lin, Spring 2024.
- Senior Capstone Project: Zachary LeClaire, Spring 2025.
- Senior Research Project: Ruoyu Shao, Spring 2025.
- Senior Capstone Project: Tina Ma, Spring 2025.

Case Western Reserve University

- Undergraduate committee and Appointments committee, MAMS, 2024-25.
- Statistics Undergraduate Sub-Committee, MAMS, 2024-25.
- Datafest organization, 2024.
- Reopening of the Statistics Ph.D. Program, 2024-25.
- Statistics Tenure-Track Faculty Search Committee, 2023-24.
- Member of Statistics Faculty Graduate sub-committee, 2023-24.

Other professional service

- Organized a session on “Advances in Bayesian Spatio-Temporal and Extreme Value Modeling” at the International Indian Statistical Association Conference, Nebraska, Lincoln, 2025.
- Organized a session on “Statistical Inverse Problems” at the International Indian Statistical Association Conference, Kochi, India, 2024.
- Panelist on “Misogyny in Academia Panel” organized by the Undergraduate Diversity Collective, Case Western Reserve University, 2024.
- Organized a session on “Statistics and its applications to Healthcare, Education and Industry” at the 50th Anniversary Celebration of School of Statistics, University of Minnesota in October 2022.
- Organizing a session on “Statistical Learning Theory” in an upcoming conference, International Indian Statistical Association (IISA) conference in December 2022.
- Session chair for the session “Contemporary Topics in Learning” at Women in Statistics and Data Science (WSDS) conference (Cincinnati 2018)
- Member of American Statistical Association.
- Refereed articles for
 - NeurIPS, ICML
 - Journal of Machine Learning Research (JMLR)
 - The American Statistician
 - Journal of the Indian Statistical Association (JISA)
 - Data Science in Science
 - Sankhya B
 - Statistical Science
 - Annals of the Institute of Statistical Mathematics
 - Journal of Statistical Research

Outreach

- (Spring 2025) Presented and interacted with members of Women’s STEM Club at Solon High School, Ohio.
- (Spring 2025) Served as Statistical consultant at the monthly Journal club, University Hospitals (UH) Psychiatry Didactics.